Capstone research report · October 2026

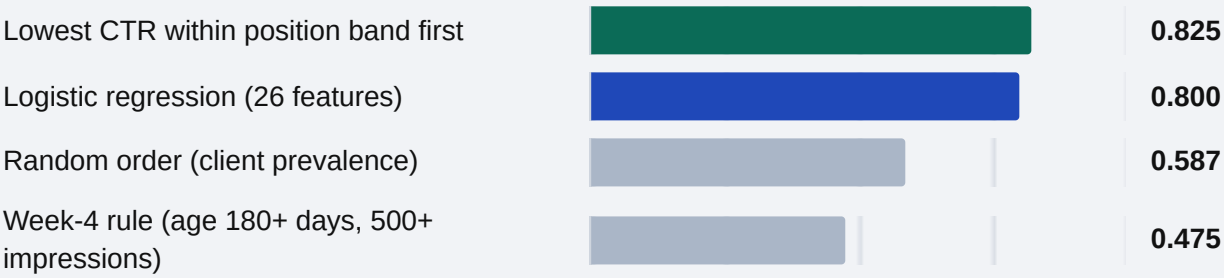
What Ranks First in a Content Review Queue?

A client-grouped test of a page-prioritization model against a rule and a one-signal baseline

By Muhammad Ahmed. Data: an anonymized FlyRank slice of 30,000 pages from 32 clients. Window: an undated trailing-90-day snapshot.

The result in one picture

Which 20 pages of one client site should a reviewer open first? A logistic regression beat a stale rule and random order inside each client's queue, but a one-line ordering, lowest CTR for the page's position first, scored as well.



Median Precision@20 across the 16 clients large enough to validate (out-of-fold, ranked pages, same-window decline proxy; higher is better). Same-window ranking, not a future forecast or causal evidence.

Takeaway: the model beats a stale rule and random order, and a one-line ordering does as well as the model.

- At K = 20 the model exceeds random order in 15 of 16 clients and the Week-4 rule in 14 of 16, and its median Precision@K is above client prevalence at every tested depth (K = 10 to 75).
- A one-signal ordering scores as well (0.825 against 0.800), and the model is strictly higher than it in only 4 of 16 clients.
- Every result is a same-window association with a proxy label. There is no forecast and no causal claim.

Reading depths. Scan: this panel. Evidence: sections 2 to 6. Audit trail: section 7, with code, seeds, environment and a result-to-notebook map.

Abstract

Which pages of one client site should a reviewer open first? This study uses an anonymized slice of 30,000 pages from 32 clients (26,612 in scope), with a same-window decline proxy as the label. A logistic regression, scored out-of-fold under client-grouped validation, is compared with a transparent rule, random order and a one-signal ordering (lowest CTR within the position band) inside each client's own queue. Across 16 validated clients the

model's median Precision@20 is 0.800 against 0.475 for the rule and 0.587 for random order, but the one-signal ordering scores as well (0.825), so the model is not shown to add value over it. The result is a human-review playbook that starts from the transparent ordering; it is a same-window ranking, not a future forecast or causal evidence.

Extended abstract

Problem. A content team cannot review every page. The decision supported here is the order in which one client's pages are reviewed, so the metric is Precision@20 inside that client's queue.

Setting. 26,612 of the 30,000 pages are in scope and 16,617 are ranked pages across 28 clients (trailing-90-day aggregates, no calendar dates). The label is a same-window *decline proxy*: impressions in the latest 30 days fell by more than 20% against the previous 30. The model is trained and scored on in-scope pages only, and validation is grouped by client.

Approach. A logistic regression on 26 page-level features is compared, on identical pages, with a transparent rule (age at least 180 days and at least 500 impressions), with random order and, as a stress test, with a one-signal ordering (lowest CTR within the page's position band first).

Evidence. Median Precision@20 is 0.800 for the model, 0.475 for the rule and 0.587 for random order, with median AUC 0.640. At $K = 20$ the model exceeds random order in 15 of 16 validated clients and the Week-4 rule in 14 of 16, and its median Precision@K is above client prevalence at every tested depth ($K = 10$, 20, 30, 50 and 75), although the share of clients above prevalence varies by depth.

The one-signal ordering scored as well (median 0.825) and the model was strictly higher than it in only 4 of 16 clients, so the model is not shown to add value over that transparent ordering on the queue's own pages. Among observable archetypes only low CTR for the page's position is consistently associated with the proxy; stale-but-visible pages declined less.

Across three reported logistic-regression evaluations with different split and preprocessing choices, pooled Precision@50 ranged from 0.46 to 0.80; these figures provide context, not an apples-to-apples comparison. A deliberately leaked field drives the score to 0.999, which shows the audit can see a leak; a random split scored above client-grouped splits on average (AUC 0.692 vs 0.641 $\pm$ 0.064 across ten seeds), by about one seed-to-seed spread.

Scope. The label window overlaps the features' window, so results are within-window associations; no calendar dates exist for a time-forward test; no causal claim about a refresh or any edit is made; and the score has not been shown to add to reading the trend in analytics. The output is a human-review playbook, not an automated decision. Code, seeds, environment and a result-to-notebook map are in section 7.

Reading guide

Each finding carries a tag so a reader can see what kind of statement it is. CONFIRMED held across clients. REVERSED the data showed the opposite of the common expectation. NUANCED partly true, with caveats. NEW ANALYSIS added in this study. NOT REPLICATED or not testable with this slice. **Precision@K** is the share of a client's top K pages that carry the decline proxy; **prevalence** is the share in the whole client, which is also what random order delivers on average. **AUC** is the chance a declining page scores above a non-declining one (0.5 is chance).

1. Introduction and claims

Problem. A content program that manages many client sites can open only a limited number of pages per review cycle. FlyRank's public data report advises reviewing aging pages before they decline [5]; this study asks a narrower question that its own slice can test.

A concrete case. Picture a reviewer with one afternoon for a client whose site holds thousands of pages (the largest client in this slice has 7,008). She can open about 20. If she opens 20 at random, about 6 in 10 of them already carry the decline proxy in this slice (a median prevalence of 0.587), so the real question is how much a ranking adds over that. This scenario is illustrative.

Research question. Can observable page-level search and engagement signals help a content team prioritize which pages to review first on one client site?

Decision supported. Which pages a reviewer opens first in a review cycle. **Who acts:** an SEO or content team working on one client at a time. **Cost of a wrong call:** review time spent on a page that needed nothing while a more useful page waits. Nothing is changed automatically, so the cost is attention, not a live edit.

Claims

Each claim carries its own boundary and points to the sections that support it.

C1 · Sections 3 and 4

Inside a client's own queue, scored out-of-fold, the model's Precision@20 is, at K = 20, above random order in 15 of 16 validated clients (median 0.800 vs 0.587) and above the Week-4 rule in 14 of 16 (median 0.475), with median AUC 0.640.

Boundary: ranked pages of 16 validated clients; a same-window decline proxy; the median model Precision@K is above client prevalence at every tested depth (K = 10, 20, 30, 50 and 75), although the share of clients above prevalence varies by depth (94% at K = 10 to 30, 100% at K = 50 and 75); deeper depths were not tested. This is not reliable superiority: across three reported logistic-regression evaluations with different split and preprocessing choices, pooled Precision@50 ranged from 0.46 to 0.80, which is context, not an apples-to-apples comparison.

C2 · Section 3

The evaluation boundary matters, but less than a single split suggested: a deliberately leaked field drives the score to 0.999, so the audit can see a leak; a random split scored above client-grouped splits on average (AUC 0.692 vs 0.641 $\pm$ 0.064; Precision@50 0.90 vs 0.808 $\pm$ 0.068 across ten seeds), by about 0.8 and 1.4 seed-to-seed SDs, and the single grouped split (AUC 0.605, Precision@50 0.62) was a low draw.

Boundary: the audit cannot remove the overlap between the feature window and the label window.

C3 · Sections 4 and 6

Among observable archetypes only low CTR for the page's position is consistently associated with the proxy (+5.6 pp, higher in 12 of 16 eligible clients); stale-but-visible pages are associated with less decline (−11.8 pp, higher in 1 of 12).

Boundary: a descriptive association inside this client mix, not a mechanism.

C4 · Section 4

A transparent one-signal ordering (lowest CTR within the page's position band first) matched the model on the queue's own pages: median Precision@20 0.825 vs 0.800 on ranked pages (a gap of half of one pick), with the model strictly higher in 4 of 16 clients. On all in-scope pages the model was ahead, 0.850 vs 0.750 (strictly higher in 10 of 21). The model is therefore not shown to add value over this ordering where the queue is used.

Boundary: the baseline was added after the verdict rule was fixed, so this is an exploratory stress test.

> **Careful words.** Same-window ranking, not a future forecast or causal evidence. A predictive feature is not a search-ranking factor, and nothing here shows that fixing a page causes recovery. This is decision support.

Related context

Baseline-first practice [1] motivates the transparent rule and the random-order reference. `GroupKFold` [2] keeps whole clients on one side of a split, and the model is a logistic regression [3] because the decision only needs an ordering a reviewer can read. Search Console data is coverage-constrained, so counts are read as reported, not as true totals [4]. No methodological novelty is claimed; the contribution is the evaluation design and its honest reading.

---

2. Data

Dataset at a glance

Measure	Value
Pages analyzed	30,000 across 32 pseudonymous clients (3 to 7,008 pages each; the largest holds 23.4% and the three largest 43.3%)
In scope (impressions in the previous 30 days)	26,612 (88.7%); proxy rate 0.611 in scope and 0.542 overall
Ranked pages (in scope, at least 500 impressions in 90 days)	16,617 across 28 clients
Validated clients (at least 100 ranked pages and 20 of each class)	16 on ranked pages; 21 on all in-scope pages
Median Precision@20 on ranked pages	Model 0.800, Week-4 rule 0.475, random order 0.587, one-signal ordering 0.825
Median AUC (IQR)	0.640 (0.607 to 0.649)
Window	Trailing 90 days to an undated snapshot; page age runs 90 to 564 days

These are counts from one anonymized slice, not a full portfolio. Per-client numbers use only clients large enough to validate, so they describe this client mix, not content pages in general.

Study scope and evidence standard

- **Release.** One anonymized starter slice: one row per content page for one of 32 pseudonymous clients. Every numeric column is a trailing-90-day aggregate ending at an undated export snapshot; the last-30-day and previous-30-day columns split the most recent 60 of those days. No page is younger than 90 days.

- **Label.** The proxy is "down": impressions in the latest 30 days fell by more than 20% against the previous 30. Rebuilding the slice's own trend column from the two 30-day impression columns matches all 30,000 rows at a $\pm$20% threshold. FlyRank's public report defines up and down at $\pm$10% [5], so the two labels are not interchangeable.
- **Time order.** Features and label come from the same snapshot, and the 90-day features contain the latest 30 days that define the label. This is same-window decline-proxy scoring: nothing here claims to predict a decline before it happens. A future claim needs earlier features and a later label, which this slice cannot supply because it has no calendar dates.
- **Scope.** Pages with no impressions in the previous 30 days cannot be "down", so the proxy is undefined for them: 3,388 pages (11.3%) are out of scope. Scope is a rule, never a feature.
- **Training scope.** The model is trained on in-scope pages only and only in-scope pages are evaluated; the 3,388 out-of-scope pages are never trained on or scored. Earlier notebooks (W03, W05, W06) trained on all pages, so their numbers are not directly comparable.
- **Evidence standard.** The headline is per-client, out-of-fold, on ranked pages. Pooled whole-fold numbers are context only. A model result never overrides a direct comparison.
- **Interpretation rule.** External SEO findings are hypotheses, not proof. Where this slice and the FlyRank report disagree, both are reported (Finding 8).

Final model features (26)

This is the only feature list in the paper; the notebook builds it once and every cell reuses it.

Numeric (18): `search_volume`, `competition`, `cpc`, `word_count`, `char_count`, `log_impressions_90d`, `log_clicks_90d`, `log_sessions_90d`, `log_ai_sessions_90d`, `days_with_impressions`, `days_with_sessions`, `content_age_days`, `days_since_last_update`, `ctr`, `avg_position`, `engagement_rate`, `scroll_rate`, `ai_traffic_pct`.

Categorical (8): `competition_level`, `content_type`, `main_intent`, `age_tier`, `freshness_tier`, `word_count_tier`, `impression_tier`, `position_tier`.

Removed fields, and why

- `trend_direction` and `trend_pct`: the label's source.
- The six last-30-day and previous-30-day columns: they rebuild the label exactly.
- `provider_used` and `model_used`: content-generation metadata, not a page signal.
- `pageviews_90d`, `users_90d`, `engaged_sessions_90d`, `scroll_events_90d`: raw counts already summarized by kept rate features, so they double-count the signal.
- `char_count_tier` and `age_tier_order`: duplicates of kept features.
- `content_id` and `client_id`: grouping and joining only.

Gotchas handled

Rate columns are percentages multiplied by 100. A position of 0 means "no data" (1,205 rows), not rank zero. Missingness is structural: `search_volume` is blank for every page of one content type and `word_count` is blank for 28.3% of another, so zero-filling would teach the model the content type. Values are imputed inside each training fold with median imputation and missingness indicators. Identifiers are pseudonymous and never printed, and no row-level predictions are written to disk.

3. Methodology

Assumptions. A page's recent change in impressions is something a team can act on, and the 90-day features describe the page well enough to rank it. The features overlap the label window, so every result is a within-window association.

Features and model. The 26 final features listed in section 2. Logistic regression with balanced class weights, chosen because it is readable and auditable. Its score ranks pages; it is not a calibrated probability.

Validation. Five-fold `GroupKFold` by client [2]: no client is in train and test together, and every client is scored once by a model that never saw it (out-of-fold). Folds are cut by client on all pages (1, 7, 8, 8 and 8 test clients; one client has no in-scope pages), and the model is trained on the in-scope pages of the training folds only. The queue is per client, so the main metrics are measured per client.

Baselines. (1) Random order, whose expected Precision@K is the prevalence. (2) The Week-4 rule: eligible if age is at least 180 days and impressions at least 500, ranked by impressions. (3) Added as a stress test: lowest CTR within the page's position band first, using the CTR percentile inside each client (the LOW_CTR_FOR_POSITION archetype instead uses the lowest global quartile within the band among ranked pages, so the two are related but not identical). It uses no model. It was added after the verdict rule was fixed and after the Week-4 rule scored at or below random, so it is exploratory.

Pre-set verdict rule. SUPPORTED if at least 8 clients are validated, and AUC is above 0.5 and Precision@20 is above prevalence in at least 70% of them.

Comparison contract: the same for the model and every baseline

Item	Setting
Eligible rows	Ranked pages: in scope (impressions in the previous 30 days) and at least 500 impressions in 90 days
Feature window	Trailing 90 days to the snapshot (contains the label window)
Label window	Latest 30 days vs previous 30 days, same snapshot (overlapping)
Folds	GroupKFold (5) by client; metrics per validated client (at least 100 ranked pages and 20 of each class)
Metric and K	Precision@20 (primary) and AUC; Precision@K for K = 10 to 75 in the depth curve
Tie rule	Model: score descending, then content ID ascending. Rule: score, impressions, older age, then content ID. One-signal ordering: score, then content ID
Baselines	Random order (equal to client prevalence); Week-4 rule; lowest CTR within position band

A baseline differs from the model only in its score.

Leakage as a method

The test must be able to see a leak before its silence means anything, so a known leak (the trend column) was added back on purpose as a test fixture; it is never a final feature. The audit also splits window features from static features, compares random with client-grouped splits, and repeats the grouped split with ten seeds so one lucky draw does not carry the result. The forbidden columns are asserted absent from the feature list.

Leakage audit on in-scope pages (supports C2); base rate 0.611

Test	AUC	P@50	Reading
Clean features, random split	0.692	0.90	Pages of the same client sit on both sides of the split
Clean + trend column (deliberate leak)	**0.999**	**1.00**	**The audit can see a leak**
90-day-window features only	0.634	0.82	About the same AUC as static features alone
Static features only	0.632	0.86	P@50 is noisy at n = 50
Clean, client-grouped 25% split (seed 42)	0.605	0.62	One draw, and a low one
Clean, client-grouped 25% split, ten seeds	**0.641 ± 0.064**	**0.808 ± 0.068**	**SD across seeds, not a confidence interval**

The random-versus-grouped AUC gap (0.05) is about 0.8 seed-to-seed SD (0.064); the Precision@50 gap (0.09) is about 1.4 SD (0.068). The audit does not remove the window overlap.

4. Results

CONFIRMED Finding 1 (within-window): inside a client's own queue, the model's top 20 held more declining pages than the rule or random order

Supports C1. Boundary: ranked pages of 16 validated clients; a same-window decline proxy; every depth tested from K = 10 to 75.

Per-client results, out-of-fold (medians across validated clients)

Scope	Clients	AUC median (IQR)	P@20 model	P@20 rule	Prevalence	Model above prevalence	Model above rule
Ranked pages (the queue)	**16**	**0.640 (0.607 to 0.649)**	**0.800**	**0.475**	**0.587**	**94%**	**88%**
All in-scope pages (sensitivity)	21	0.610 (0.574 to 0.645)	0.850	0.450	0.644	90%	86%

Verdict under the pre-set rule: SUPPORTED in both runs. AUC was above 0.5 in every validated client of the queue run (90% in the sensitivity run). The rule is below random order (0.475 against 0.587), so the fairer yardstick is random: the model gains 21.3 percentage points over random order and 32.5 over the rule, and the rule is a weak baseline.

Where the advantage stops: Precision@K, median across validated clients with at least 2K ranked pages

K	Clients	Model	Week-4 rule	Prevalence	Model above prevalence
10	16	0.800	0.400	0.587	94%
20	16	0.800	0.475	0.587	94%
30	16	0.800	0.467	0.587	94%
50	16	0.810	0.490	0.587	100%
75	13	0.800	0.507	0.622	100%

The model's median stays above prevalence at every depth tested, in 94% of clients at K = 10 to 30 and in all clients at K = 50 and 75. Deeper depths were not tested.

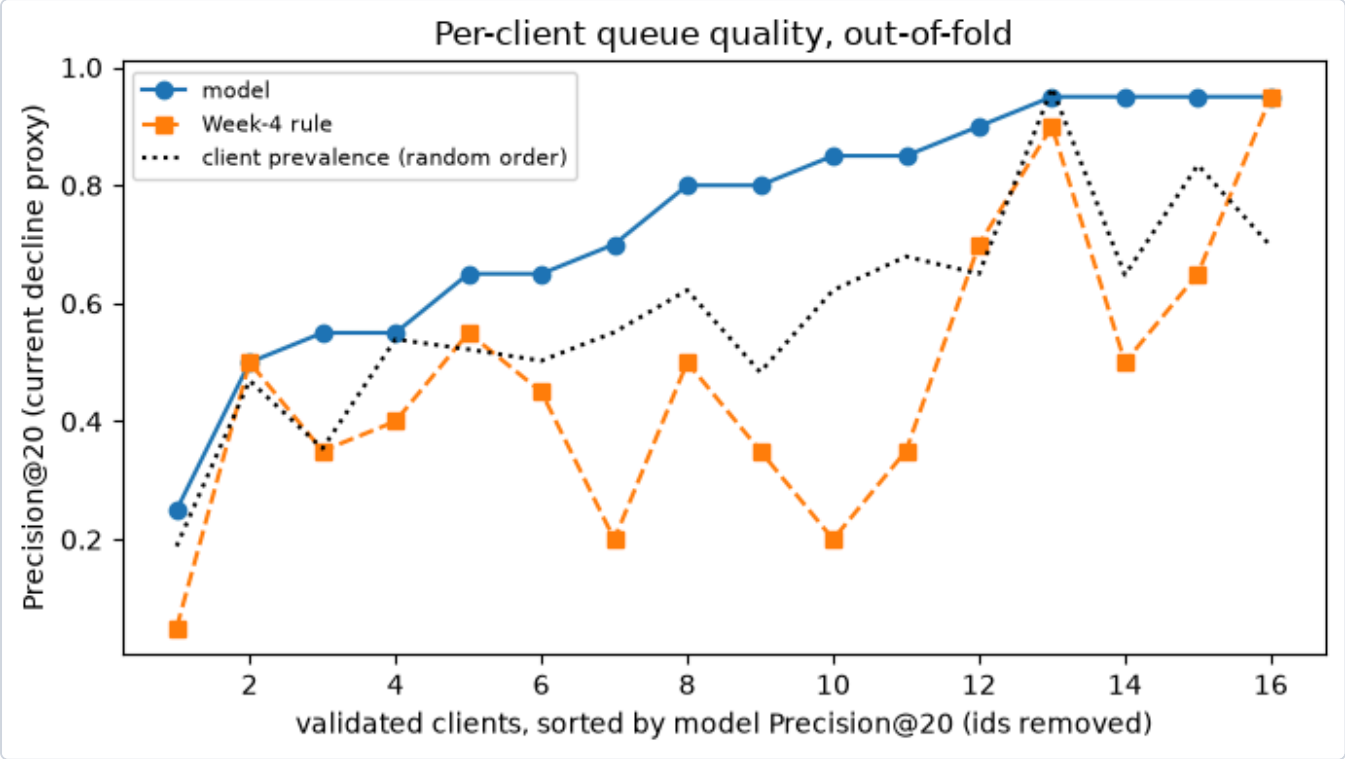


Figure 3. Per-client Precision@20, out-of-fold, ids removed. Each point is one client; the model sits above prevalence in 15 of 16. It supports C1 only. It does not show the one-signal ordering (Finding 2).

Takeaway: the model sits above client prevalence in 15 of 16 clients, with a median of 0.800 against 0.587.

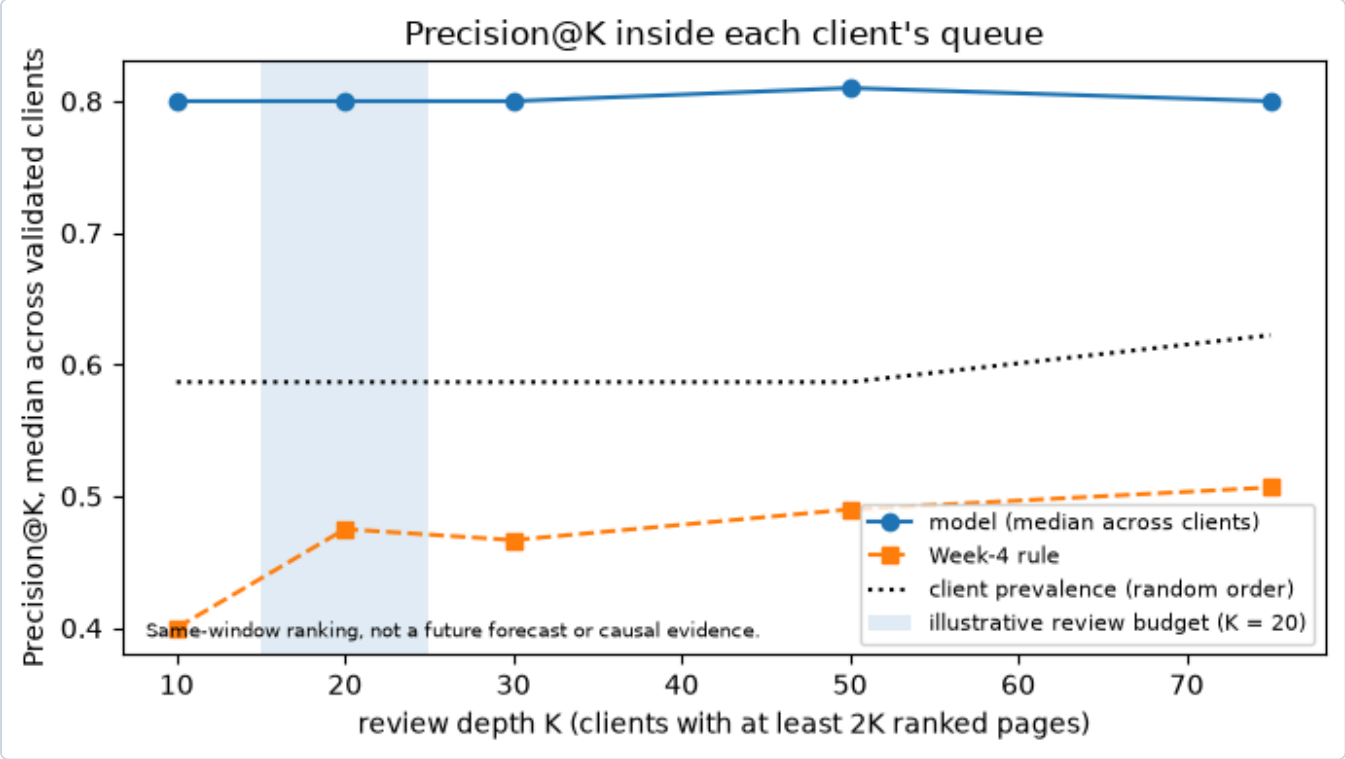


Figure 4. Median Precision@K across validated clients. It shows where the model's advantage over prevalence sits (everywhere tested) and supports the boundary of C1. The one-signal ordering is not plotted; its median Precision@20 is in Finding 2. The shaded band is an illustrative review budget of 20.

Takeaway: the model's median Precision@K stays near 0.80 from K = 10 to 75, above prevalence at every depth tested.

NUANCED Finding 2: a one-signal ordering matched the model on the queue's own pages

Supports C4. Boundary: the baseline was added after the verdict rule was fixed; one stress-test run.

Stress test: lowest CTR within the position band first vs the model

Scope	Clients	P@20 model	P@20 one-signal	P@20 rule	Prevalence	Model strictly higher than one-signal	One-signal above prevalence
Ranked pages (the queue)	**16**	**0.800**	**0.825**	**0.475**	**0.587**	**25% (4 of 16)**	**94%**
All in-scope pages	21	0.850	0.750	0.450	0.644	48% (10 of 21)	76%

On the queue's own population the transparent ordering scores slightly higher than the model: a median gap of 2.5 points, which is half of one pick at K = 20. Across all in-scope pages the model is ahead by 10 points. The evidence is mixed outside the queue and a near-tie inside it.

The reading is that the signal, low CTR for a page's position, carries most of the value, and the model adds little over it where the queue is used. A caution applies: CTR is clicks divided by impressions over 90 days, so a burst of impressions that has since ended can produce both a low CTR and a later decline. That mechanical reading was not tested; if it holds, the ordering detects finished impression bursts rather than pages that need a snippet fix.

NUANCED Finding 3: pooled whole-fold numbers look bigger and swing with the held-out clients

Context for C1 and C2; not a claim by itself. In-scope pages only.

Whole folds pooled across clients (context only)

Fold	Held-out clients	In-scope pages	Prevalence	Model P@50	Rule P@50	AUC	Largest client in model top 50	Largest client in rule top 50
1	1	6,983	0.492	0.80	0.40	0.641	1.00	1.00
2	7	5,557	0.666	0.86	0.48	0.644	0.78	1.00
3	7	4,432	0.493	0.66	0.38	0.648	0.46	1.00
4	8	4,857	0.737	0.78	0.60	0.494	0.50	0.82
5	8	4,783	0.703	0.88	0.84	0.548	0.58	0.94
Mean			**0.618**	**0.796**	**0.540**	**0.595**		

Fold 1 is a single client, and one client with no in-scope pages sits in fold 3. In fold 4 the pooled AUC (0.494) is below chance while model Precision@50 (0.78) is above prevalence (0.737), so an advantage at the top of the list can coexist with no overall ranking skill. In fold 5 the rule (0.84) is close to the model (0.88).

Three reported logistic-regression evaluations (different split and preprocessing choices)

Evaluation	Pages used	Held-out clients	P@50 model	P@50 rule	Prevalence	AUC
Grouped 25% split, seed 42 (earlier notebook, W03)	all pages	about a quarter of clients	0.74	not computed	not computed	0.611
Single holdout (earlier notebook, W05)	all pages	6 of 32	0.46	0.30	0.391	0.700
Five-fold mean (this study)	in-scope pages	all 31 with in-scope pages, in turn	0.796	0.540	0.618	0.595

Pooled Precision@50 ranged from 0.46 to 0.80 across the three evaluations. These figures provide context, not an apples-to-apples comparison: the earlier notebooks used all pages with a zero-filled pipeline, while this study trains and scores on in-scope pages only.

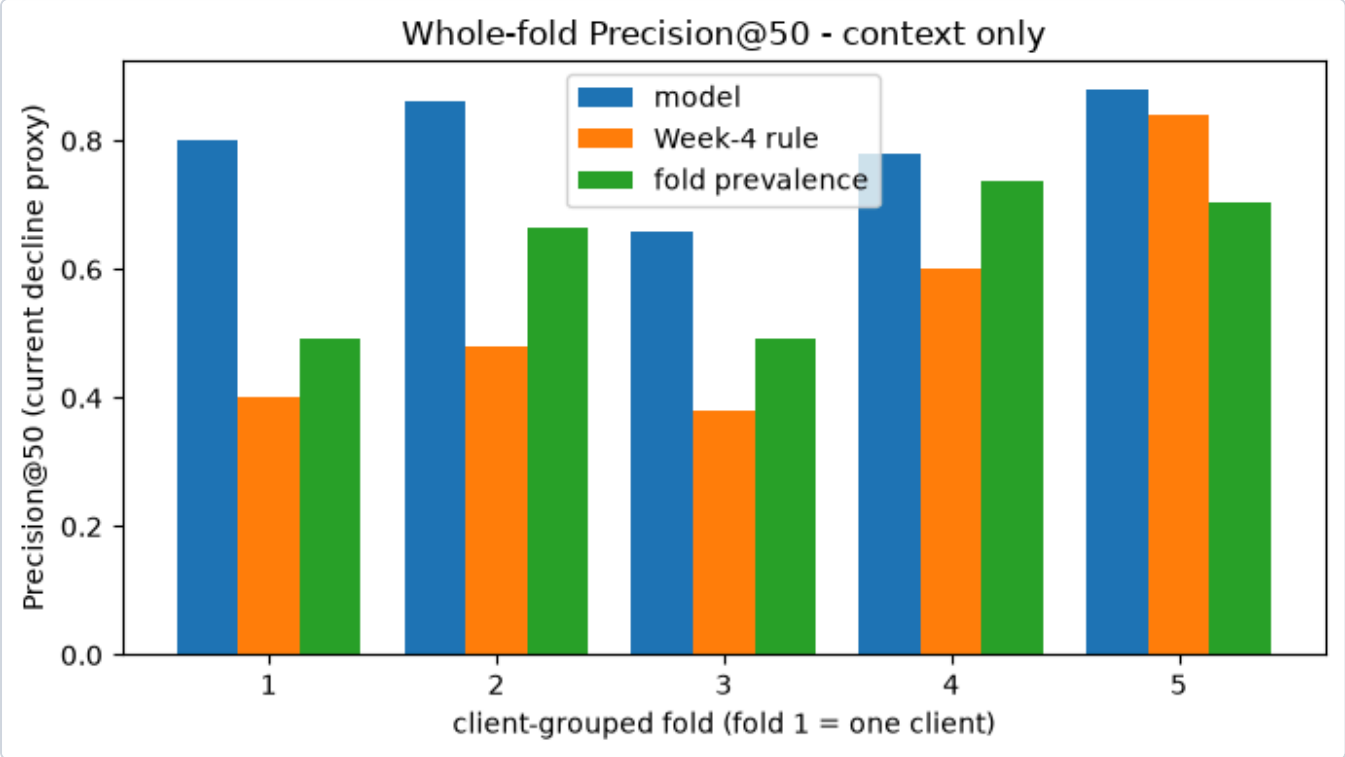


Figure 1. Whole-fold Precision@50, in-scope pages. Compare bars within a fold, not across folds, because fold prevalence changes the baseline. Context for C1; it is not the queue's performance.

Takeaway: pooled Precision@50 for the model swings from 0.66 to 0.88 across folds, so read it as context, not as the queue's performance.

CONFIRMED Finding 4 (one signal): low CTR for the page's position is the only archetype consistently associated with the proxy

Supports C3. Boundary: a descriptive association inside this client mix, not a mechanism.

Low CTR means the lowest 25% of CTR within the page's position band, taken across all ranked pages: (0,3], (3,5], (5,10], (10,20] and (20,+). HIGHER or LOWER requires a gap of at least 5 percentage points in the same direction in at least 70% of at least five eligible clients (30 or more pages inside and outside the group). The one-signal ordering in Finding 2 is related but not identical: it uses the CTR percentile inside each client.

Archetype vs the rest of the in-scope pages, compared inside each client

Archetype	Pages	Decline rate	Rest	Gap (pp)	Eligible clients	Higher in	Evidence
LOW_CTR_FOR_POSITION	**4,470**	**0.657**	**0.602**	**+5.6**	**16**	**75%**	**HIGHER_THAN_REST**
VISIBLE_OTHER	5,052	0.653	0.601	+5.2	10	30%	MIXED
STALE_AND_VISIBLE	7,095	0.524	0.643	−11.8	12	8%	LOWER_THAN_REST
LOW_VOLUME (monitor only)	9,995	0.630	0.599	+3.1	18	78%	MIXED

Descriptive; no confidence interval. "Higher in" is the share of eligible clients where the group declined more than the rest of that client.

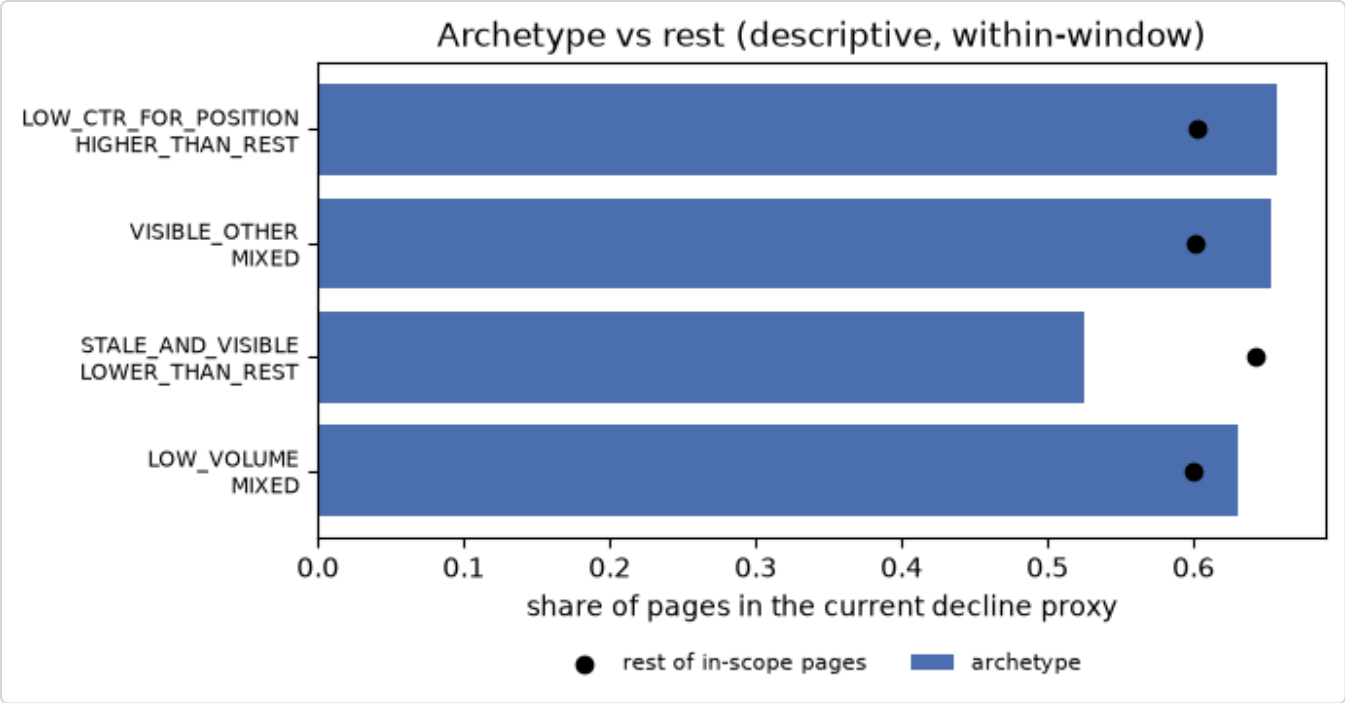


Figure 2. Decline-proxy rate by archetype (bars) against the rest of the in-scope pages (dots). A gap is an association with the proxy, compared inside clients. It supports C3 and does not show that any feature is a ranking factor or that fixing a page causes recovery.

Takeaway: low-CTR pages decline more than the rest (0.657 vs 0.602), and stale-but-visible pages decline less (0.524 vs 0.643).

REVERSED Finding 5: old and stale pages did not decline more

Supports C3.

Age and update comparisons, inside clients

Comparison	Pages (group vs rest)	Decline rate	Eligible clients	Higher in	Reading
Older than 365 days	6,107 vs 20,505	0.444 vs 0.661	7	29%	Declined LESS
Younger than 180 days	10,675 vs 15,937	0.705 vs 0.548	10	80%	Declined MORE
Updated within 30 days	17,454 vs 9,158	0.600 vs 0.632	12	42%	No consistent pattern

The update column is not a refresh event: its most common values are 20 (36% of in-scope pages), 104 (32%) and 22 (13%), values of 30 or less fall inside the label window, and there is no pre-update baseline. Age is confounded with client, so these comparisons rest on few clients and are tentative. A young page's decline may reflect a launch or ramp pattern rather than decay.

NUANCED Finding 6: the top of the queue is dominated by young, zero-engagement pages, and a third of it comes from unvalidated clients

Supports the boundary of C1 and the caution flags in section 6.

Caution flags: ranked pages vs the top-20-per-client rows (478 rows)

Caution	Ranked pages	Top-20 rows
AGE_UNDER_180D	40.3%	58.8%
ZERO_ENGAGEMENT	55.4%	77.6%
TOP1PCT_IMPRESSIONS	1.0%	0.4%
POSITION_MISSING	0.0%	0.0%
UNVALIDATED_CLIENT	1.7%	33.1%

Among all in-scope pages 40.1% are under 180 days old and 69.3% have zero engagement. A flag this common describes the typical page, not a per-page warning. The top-20 rows are 307 low-CTR, 88 stale-and-visible and 83 other visible pages. 158 of the 478 rows come from queued clients without their own validation. Real decline, a launch ramp and a tracking gap cannot be separated with this slice.

NEW ANALYSIS

Finding 7 (illustrative): review yield per client per cycle

Supports the playbook in section 6; not a claim.

Placeholders: 15 minutes per page and 40 per hour. “Found” means the page carries the proxy, not that a fix will recover it.

Mean yield across the 16 validated clients

Slots	Method	Mean yield	Pages found	Review hours	Cost per page found
20	Model queue	0.741	14.8	5.0	13.50
20	Week-4 rule	0.475	9.5	5.0	21.05
20	Random order	0.583	11.7	5.0	17.15
50	Model queue	0.724	36.2	12.5	13.82
50	Week-4 rule	0.498	24.9	12.5	20.10
50	Random order	0.583	29.1	12.5	17.15

There is no row for the one-signal ordering; its median Precision@20 (0.825) is in Finding 2. Break-even is not computed: it needs the action success rate and the value of a recovered page, and neither is in the data.

NOT REPLICATED

Finding 8: cross-check with the FlyRank March 2026 report (57 brands, 341,701 pieces)

A cross-check, not a claim of this study.

Differences in label (±10% vs ±20%), scope (this slice requires a previous-30-day baseline) and population mean none of these rows is a refutation.

FlyRank report vs this slice

FlyRank report [5]	This slice	Reading
Growing pages are younger (184 vs 230 days)	Pages under 180 days declined more (0.705 vs 0.548); pages over 365 days declined less (0.444 vs 0.661)	NOT REPLICATED
Health falls to 14 at 271 to 365 days	No health score; age starts at 90 days and is not banded the same way	NOT TESTED
Refreshed 365+ day pages: 3.2x health, 57x impressions	No refresh-event column; the update column is not a refresh event	NOT TESTABLE

5. Limitations

What this work cannot claim, and which claim each limit protects.

- **No future prediction (C1, C2).** The label is a current-window proxy and the 90-day features contain the label's own window. No calendar dates exist, so no later test period can be built. A future claim needs earlier features and a later label.
- **Same-window ranking, not a future forecast or causal evidence (C3).** A predictive feature is not a search-ranking factor, and the archetype gaps are associations, not mechanisms.
- **No refresh claim (C3).** No dated refresh events, no pre-refresh baseline, no matched control group and no seasonality control exist; the update column is not a refresh event.
- **The label can be read directly (C1).** The label is the 30-day impression change (Spearman 1.000 with the trend column), which a team can read in analytics. Whether the score adds anything beyond reading that trend was not tested.
- **Low CTR may be mechanical (C3, C4).** A burst of impressions that has since ended can produce both a low CTR and a later decline. That reading was not tested.
- **A baseline added late (C4).** The one-signal ordering was added after the verdict rule was fixed and after the Week-4 rule scored at or below random; it is one exploratory stress-test run, and its median gap to the model (2.5 points) is half of one pick at K = 20.
- **Modest, client-dependent discrimination (C1).** Median per-client AUC is 0.640; pooled whole-fold AUC ranged from 0.494 to 0.648 and was below chance in fold 4 while Precision@50 there was above prevalence, so an advantage at the top of the list can coexist with no overall ranking skill. Pooled Precision@50 ranged from 0.46 to 0.80 across three reported evaluations with different split and preprocessing choices. The advantage over prevalence held at every depth tested (K = 10 to 75); deeper depths were not tested.
- **The audit has a boundary (C2).** The leak harness, the window/static split and the grouped repeats do not remove the overlap between the feature window and the label window. Across ten seeds the random-versus-grouped AUC gap (0.05) is smaller than the seed-to-seed SD (0.064), so the grouped boundary mattered less here than a single split suggested.
- **Coverage (C1).** 3,388 pages are out of scope, 9,995 low-volume pages are monitor-only, and only 16 of the 28 queued clients are large enough for their own validation. The other 12 carry the UNVALIDATED_CLIENT flag and supply 158 of the 478 top-20 rows (33.1%). The results describe this client mix, not content pages in general.
- **Young, low-engagement pages (C1, C3).** They are over-represented in the top of the queue; real decline, a launch ramp and a tracking gap cannot be separated here.
- **Training scope.** Training and scoring use in-scope pages only; the 3,388 out-of-scope pages are never trained on or scored. Earlier notebooks trained on all pages, so their numbers are not comparable.

- **Not tested.** Non-linear models and feature interactions, feature importance, calibration, a time-aware split, tier-column provenance, whether missingness indicators partly encode client identity, feature-group ablation, small-client behaviour (they fall below the validation size rule), and whether any review prompt actually helps.
- **Data.** One cached snapshot, no calendar dates, 1,205 positions coded as no data, structurally missing fields, and uncalibrated scores from different fold models that are not comparable across clients.

---

6. Ranked recommendations

Ranked by how consistently each pattern held across clients and how cheaply it can be checked. Every "Expected" line is a hypothesis: this study tested ranking, not edits.

The queue. One queue per client. The queue built in the notebook is ordered by the model's out-of-fold score (ties by content ID): the top 20 per client are P1 (review first), the next are P2, and low-volume pages (under 500 impressions) are listed as monitor-only with no rank. There is no pooled ranking across clients. Because a transparent one-signal ordering matched the model on ranked pages (C4), a deployment should start from that ordering and keep the model score as a second view.

1. Start from the transparent ordering and keep the model as a second view

- **Why:** lowest CTR within the position band had a median Precision@20 of 0.825 against the model's 0.800 on ranked pages (C4), with no model to maintain.
- **How:** rank each client's ranked pages by CTR percentile within the position band, review the top 20, and show the model score beside each row.
- **Expected (hypothesis):** similar or better precision of the proxy with nothing to retrain.
- **Measure:** within-client Precision@20 against that client's prevalence, for both orderings, each cycle once labels arrive.

2. Review low-CTR-for-position pages with a snippet check

- **Why:** the only archetype above the rest across clients (0.657 vs 0.602, +5.6 pp, higher in 12 of 16 eligible clients).
- **How:** compare the title, meta description and snippet with the search intent and with nearby ranking pages; edit only after a reviewer approves.
- **Expected (hypothesis):** better click capture on pages already losing impressions. No edit was tested here.
- **Measure:** clicks, CTR and position 30 and 60 days after an edit, against comparable pages left alone.

3. Confirm the decline in analytics before touching a page

- **Why:** the label is the 30-day change in impressions, which a team can read directly.
- **How:** open the page's 90-day trend; check seasonality, tracking and launch date.
- **Expected (hypothesis):** fewer reviews spent on a tracking gap or a launch ramp.
- **Measure:** reviewer decisions (act, defer, reject) per cycle.

4. Treat stale-but-visible pages as a freshness check, not a decay diagnosis

- **Why:** this archetype declined less than the rest (−11.8 pp, higher in 1 of 12 clients), and pages over 365 days declined less.
- **How:** check whether facts, examples and sources are out of date, then decide update, consolidate or leave.
- **Expected (hypothesis):** none claimed; this is an editorial prompt, not a predicted gain.
- **Measure:** record the decision and revisit after a dated refresh test (action 7).

5. Run a data check on young, zero-engagement and unvalidated-client rows first

- **Why:** 58.8% of top-20 rows are under 180 days old, 77.6% have zero engagement and 33.1% come from unvalidated clients.
- **How:** once per client and page type, check engagement tracking and launch dates; review UNVALIDATED_CLIENT rows manually.
- **Expected (hypothesis):** separates real decline from ramp patterns and tracking gaps.
- **Measure:** the share of top-20 rows rejected after the data check.

6. Keep queues per client and start where the ordering can be validated

- **Why:** pooled Precision@50 ranged from 0.46 to 0.80 across three reported evaluations with different splits and preprocessing, and pooled queues are dominated by one or two clients.
- **How:** rank inside a client only; start with the 16 clients that meet the validation size rule.
- **Expected (hypothesis):** more stable queues.
- **Measure:** within-client Precision@20 against prevalence each cycle.

7. Run a real refresh test before claiming a refresh helps

- **Why:** this slice has no dated refresh events, so the question cannot be answered here.
- **How:** log each refresh date; keep a pre-refresh baseline; keep a matched untouched control group; follow up at 30, 60 and 90 days; control for seasonality.
- **Expected (hypothesis):** an answer, either way.
- **Measure:** impressions, clicks and position for refreshed vs control pages.

Archetype to action map

Priority	Archetype	Evidence	Review prompt
1	LOW_CTR_FOR_POSITION	HIGHER_THAN_REST	Check title, meta description and snippet against search intent and nearby ranking pages
2	VISIBLE_OTHER	MIXED	Quick look only if capacity remains
3	STALE_AND_VISIBLE	LOWER_THAN_REST	Check freshness; decide update, consolidate or leave (not a decay diagnosis)
4	LOW_VOLUME	MIXED	Monitor only (no review slot, no rank)

Human review

Before acting on any row, a reviewer confirms the decline in analytics; checks the caution flags (top-1% impressions, under 180 days old, missing position, zero engagement, unvalidated client; the common ones are checked once per client and page type); checks the page's business role (brand, hub, legal and pricing pages need an owner); gets a second reviewer for top-1% impression pages; and records act, defer or reject. Rows flagged UNVALIDATED_CLIENT (158 of the 478 top-20 rows, 33.1%) get a manual check first. The queue helps a reviewer decide what to look at; it never authorizes a rewrite.

The no-go list, with where each rule is enforced

ASSERTED_IN_NOTEBOOK means an assert in the notebook backs it; ENFORCED_UPSTREAM means it is enforced outside the notebook; PROPOSED means it is policy that is not implemented yet.

No-go list with enforcement status

Policy	Status	Where
Ranks are per client; no pooled cross-client ranking	ASSERTED_IN_NOTEBOOK	Assert on the rank column
Low-volume and out-of-scope pages get no rank	ASSERTED_IN_NOTEBOOK	Assert on the ranked frame
A page with no previous-30-day impressions is never labelled down	ASSERTED_IN_NOTEBOOK	Assert in the data cell
Label-source columns never enter the features	ASSERTED_IN_NOTEBOOK	Assert in the methodology cell
Only in-scope pages are used to train the model	ASSERTED_IN_NOTEBOOK	Assert in the fold loop
No client crosses train and test	ASSERTED_IN_NOTEBOOK	Assert in the fold loop
Rows of unvalidated clients carry UNVALIDATED_CLIENT	ASSERTED_IN_NOTEBOOK	Flag uses the same size rule as the validation
Identifiers are pseudonymous; no names or URLs in outputs	ENFORCED_UPSTREAM	Source release and data contract; the notebook prints no ids
Row-level queue and predictions stay out of git	ENFORCED_UPSTREAM	Repo ignore rules; the notebook writes aggregates only
No automatic publish, edit, redirect, consolidate or delete from a queue row	PROPOSED	No integration exists here; a deployment needs an approval gate
The score is never read as a probability or an expected gain; no "will recover" or "caused" wording	PROPOSED	Wording policy for reviewers and reports
No retraining or threshold change without a person reading the monitoring report	PROPOSED	No retraining code exists here
Unvalidated clients are reviewed manually until validated	PROPOSED	Needs a deployment-time rule

Monitoring on two clocks

Pre-release checks need no labels and run before a queue goes out. Post-label checks need outcomes or reviewer decisions; they can change the next queue but cannot change what was already reviewed. The thresholds are untuned starting points, not a shipped system, and a fired trigger means a person pauses and looks.

Monitoring triggers

Clock	Trigger (starting threshold)	Status
PRE_RELEASE	Input drift: missing share of a key input rises by more than 5 points	PROPOSED
PRE_RELEASE	Score drift: PSI above 0.2 for a client against its own reference	HARNESS_IN_W07_NOTEBOOK_ONLY
PRE_RELEASE	Scope drift: share of pages with previous-30-day impressions moves by more than 10 points	PROPOSED
PRE_RELEASE	Queue composition: share of flagged top-20 rows moves by more than 10 points	PROPOSED
PRE_RELEASE	New client, schema change or retrained model: re-run the leakage audit and grouped validation	PROPOSED
POST_LABEL	Reviewer rejection: above 50% of top-20 rows for two consecutive cycles	PROPOSED
POST_LABEL	Lost lift: no within-client lift for two consecutive cycles	PROPOSED
BOTH	Scheduled review every quarter	PROPOSED

The PSI harness exists only in an earlier notebook (W07: a reference for 19 of the 31 clients with in-scope pages; in its smoke test on 17 clients random-half PSI had a median of 0.010 and a maximum of 0.074, while a truncated batch scored at least 5.153). It is not in the capstone notebook and not in any production queue.

> **Fall-back rule (proposed).** If the model's within-client Precision@20 does not beat the transparent one-signal ordering for two consecutive labelled cycles, use the transparent ordering and stop maintaining the model. On this evidence the model did not beat it on ranked pages (0.800 vs 0.825).

Same-window ranking, not a future forecast or causal evidence. We observed, we measured, we support decisions. We made no causal claim.

---

7. Reproducibility

Every number in this paper comes from one notebook, `capstone.ipynb`, which reruns end to end from a fresh clone and prints a number-by-number match check against this text.

Environment, data and seeds

```
python         3.11.0
numpy          2.4.6
pandas         3.0.3
scikit-learn   1.9.0
matplotlib     3.11.1
data           data/raw/content_refresh_anonymized.csv
               30,000 rows, 44 columns, SHA-256 prefix 99a2bdada7d15ce0
seeds          42 for the random split, the grouped 25% split and the logistic regression
               0 to 9 for the grouped-split repeats
               GroupKFold is deterministic and takes no seed
```

Run

```
# fresh clone of the public repository
pip install -r requirements.txt
# run the notebooks in card order: w01 to w07, then capstone
# Colab: Runtime, Run all
# local:  jupyter nbconvert --to notebook --execute work/notebooks/capstone.ipynb
```

Result-to-notebook map

Result	Regenerated by
Dataset at a glance, final features, removed fields	`capstone.ipynb` data cell
Leakage audit and ten-seed repeat (C2)	`capstone.ipynb` methodology cell
Findings 1 to 3, comparison contract, depth curve (C1, C4)	`capstone.ipynb` results cell
Findings 4 to 7, caution flags, queue (C3)	`capstone.ipynb` recommendations cell
No-go list and monitoring tables	`capstone.ipynb` recommendations cell
Figures 1 to 4 and the metrics file	`capstone.ipynb` artifacts cell
Match check of every quoted number	Last block of the artifacts cell, and the extended-checks cell at the end of the notebook
Quoted earlier values (grouped 25% split, six-client holdout, PSI smoke test)	`w03_feature_leakage_check.ipynb`, `w05_model.ipynb`, `w07_action_playbook.ipynb`

Figures and files

Artifact	What it shows	Supports
Figure 1: Precision@50 by fold	Model, rule and prevalence per client-grouped fold; fold 1 is a single client	Context for C1
Figure 2: archetype vs rest	Decline-proxy rate per archetype with its evidence tag	C3
Figure 3: per-client Precision@20	Model, rule and prevalence for each validated client, ids removed	C1
Figure 4: Precision@K curve	Median Precision@K for K = 10 to 75 with a “same-window ranking, not a future forecast or causal evidence” note on the chart	Boundary of C1
Metrics file	Aggregates only; no row-level queue, no ids, no out-of-fold predictions	All

Every number quoted in this paper was taken from the executed run of the final notebook. Two check blocks (the end of the artifacts cell and the extended-checks cell) compare each quoted number with the run and print OK or DIFF; in the executed run both blocks printed OK for every check (53 of 53 and 139 of 139). Source repository: FlyRank-Internship-ML on GitHub; the capstone notebook is `work/notebooks/capstone.ipynb` and the weekly notebooks are in `work/notebooks`; aggregates in `work/outputs/capstone_metrics.json`.

Acknowledgments & data credit

Data: an anonymized content-performance slice provided by **FlyRank** (flyrank.ai), used here for learning and research under the internship. Data credit thanks the source; it does not replace the citations below. Client identifiers are pseudonymous, and no client names, URLs or queries appear in this report. FlyRank’s report “The State of AI-Driven SEO” (March 2026) was used as the format reference and for the cross-check in Finding 8; none of its figures are presented as results of this study. Thanks to my reviewers for pushing on leakage, split design and careful wording.

Document and data integrity statement

Every figure in this report is quoted from the executed capstone notebook and, where labelled as quoted, from earlier weekly notebooks. The slice is a cached snapshot with no calendar dates, so all results are within-window associations with a proxy label and are offered as decision-support evidence, not as guarantees.

References

1. Google. Rules of Machine Learning (https://developers.google.com/machine-learning/guides/rules-of-ml).
2. scikit-learn. GroupKFold (https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GroupKFold.html).
3. scikit-learn. LogisticRegression (https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html).
4. Google Search Console Help. About Search Console data (https://support.google.com/webmasters/answer/96568?hl=en).
5. FlyRank. The State of AI-Driven SEO, March 2026. flyrank.ai (https://flyrank.ai).

Capstone research report, October 2026. Decision support: same-window ranking, not a future forecast or causal evidence. Data credit: FlyRank (flyrank.ai).